

B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

Third & Fourth Semester

21CSC202J - OPERATING SYSTEMS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Answer all Questions

Marks BL CO

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|---|---|---|---|---|
| 1. Which among the following is the fastest among all the categories in a memory hierarchy ?
(A) Main memory
(C) CPU registers | (B) L1 Cache memory
(D) Secondary Memory | 1 | 2 | 1 |
| 2. Device controller informs CPU that it has finished its operation by causing _____.
(A) Interrupt
(C) Information | (B) Message
(D) Command | 1 | 1 | 1 |
| 3. Which of the following system calls is used for process creation in Unix-based operating systems?
(A) exec()
(C) wait() | (B) fork()
(D) kill() | 1 | 1 | 1 |
| 4. Which of the following memory management techniques does an operating system use to ensure efficient memory utilization?
(A) Paging and Segmentation
(C) Swapping and Pipelining | (B) Virtualization and Scheduling
(D) Multithreading and Synchronization | 1 | 2 | 1 |
| 5. A process can be _____.
(A) Both single threaded and multithreaded
(C) Single threaded | (B) Multithreaded
(D) Stack | 1 | 1 | 2 |
| 6. Systems that permit only a single process to execute at any given time are referred to as _____.
(A) Uniprogramming systems
(C) Unitasking systems | (B) Uniprocessing systems
(D) Multiprocessing systems | 1 | 1 | 2 |
| 7. Which synchronization mechanism allows multiple processes to access a resource while ensuring mutual exclusion?
(A) Semaphore
(C) Context Switching | (B) Deadlock
(D) Paging | 1 | 2 | 2 |
| 8. In Interprocess Communication (IPC), which of the following methods allows processes to communicate without using shared memory?
(A) Pipes
(C) Mutex Locks | (B) Semaphores
(D) Threads | 1 | 2 | 2 |
| 9. Which of the following is NOT a CPU scheduling algorithm?
(A) First-Come, First-Served (FCFS)
(C) Shortest Remaining Time Next (SRTN) | (B) Round Robin (RR)
(D) Least Recently Used (LRU) | 1 | 1 | 3 |
| 10. Select the type of CPU Scheduling algorithm that has a large average waiting time? | | | | |

(A) First come First Serve (FCFS)	(B) Priority based Scheduling Algorithm	1	1	3
(C) Round Robin Scheduling Algorithm (RR)	(D) Multilevel Feedback Queue Algorithm			
11. Which deadlock handling method involves ensuring that at least one of the necessary conditions for deadlock cannot hold?		1	2	3
(A) Deadlock Prevention	(B) Deadlock Avoidance			
(C) Deadlock Detection	(D) Recovery from Deadlock			
12. Identify the deadlock handling technique in the Banker's Algorithm.		1	2	3
(A) Deadlock Prevention	(B) Deadlock Avoidance			
(C) Deadlock Detection and Recovery	(D) Deadlock Ignorance			
13. In a paging system, what is the primary function of a page table?		1	1	4
(A) To store executable code of a process	(B) To translate logical addresses to physical addresses			
(C) To allocate contiguous memory blocks	(D) To manage file system permissions			
14. Which of the following is not a form of memory?		1	1	4
(A) Instruction cache	(B) Instruction register			
(C) Instruction opcode	(D) Translation lookaside buffer			
15. In RAID (Redundant Array of Independent Disks), which level provides data mirroring for fault tolerance?		1	2	4
(A) RAID 0	(B) RAID 1			
(C) RAID 5	(D) RAID 6			
16. The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by		1	2	4
(A) The instruction set architecture	(B) Page size			
(C) Physical memory size	(D) Number of processes in memory			
17. Which of the following is NOT a common type of program threat?		1	1	5
(A) Trojan Horse	(B) Denial-of-Service (DoS) Attack			
(C) Virus	(D) Worm			
18. _____ is an approach to restricting system access to authorized users.		1	1	5
(A) Role-based access control	(B) Process-based access control			
(C) Job-based access control	(D) None of the mentioned			
19. What is the primary goal of protection in an operating system?		1	2	5
(A) To prevent hardware failures	(B) To ensure that system resources are used only according to policies			
(C) To optimize CPU scheduling algorithms	(D) To speed up program execution			
20. Which principle states that programs, users and even the systems be given just enough privileges to perform their task?		1	2	5
(A) Principle of operating system	(B) Principle of least privilege			
(C) Principle of process scheduling	(D) None of the mentioned			

PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

Marks BL CO

- | | | | | |
|-----|--|---|---|---|
| 21. | (a) Describe in detail the various services provided by an operating system, explaining how each service contributes to the overall functionality and performance of the system. | 8 | 3 | 1 |
|-----|--|---|---|---|
- (OR)**
- | | | | | |
|--|--|--|--|--|
| | (b) Write a note on Monolithic, Layered and microkernel OS structure and discuss pros and cons for the same. | | | |
|--|--|--|--|--|

22. (a) Explain the Process Control Block (PCB) and its significance in process management. What are the key attributes stored in a PCB? 8 3 2
(OR)
(b) What is the Critical-Section Problem? Explain Peterson's Solution and how it ensures mutual exclusion in process synchronization.
23. (a) Discuss the concept of real-time CPU scheduling. How do real-time scheduling algorithms differ from traditional scheduling algorithms, and what challenges are involved in real-time scheduling? 8 3 3
(OR)
(b) What are the necessary conditions for a deadlock to occur? Explain different methods for handling deadlocks, including prevention, avoidance, and detection.
24. (a) Explain the concept of paging and segmentation in memory management. What are their advantages and disadvantages? 8 3 4
(OR)
(b) Describe different disk scheduling algorithms used in operating systems.
25. (a) Explain the goals and principles of protection in an operating system. How does protection help in ensuring system security and efficient resource utilization? 8 3 5
(OR)
(b) Explain with example Threats to Protection and Security.

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

Marks BL CO

26. Consider four processes with the length of CPU burst time, Arrival time and Priority (small value means high priority) 15 3 3

PROCESS	BURST TIME	ARRIVAL TIME	PRIORITY TIME
P1	40	0	2
P2	55	15	4
P3	35	15	1
P4	10	35	3

- (i) Create Gantt charts illustrating the execution of these processes using FCFS, non preemptive SJF, priority scheduling and Round Robin scheduling (quantum = 5 ms) algorithms for this set of processes.
(ii) Calculate the turnaround time for process P3 under each of the above algorithms.
(iii) Calculate average waiting time for each algorithm and find out which algorithm would give the minimum average waiting time.
27. A disk has 200 cylinders, numbered from 0 to 199. The disk head is currently at cylinder 100. The following requests are in the queue: 15 3 4
- Request 1: Cylinder 50
- Request 2: Cylinder 120
- Request 3: Cylinder 30
- Request 4: Cylinder 180
- Request 5: Cylinder 70
Using each of the following disk scheduling algorithms, determine the order in which the requests are serviced and the total distance traveled by the disk head:
1. First-Come-First-Served (FCFS)
2. Shortest-Seek-Time-First (SSTF)
3. SCAN (Elevator Algorithm)
4. C-SCAN (Circular SCAN)
5. LOOK (Circular LOOK)
6. C-LOOK (Circular LOOK).

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